

RESEARCH ARTICLE

Determination of air quality impacts of a cement production facility in Düzce province using AERMOD air quality dispersion model

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ABSTRACT

In this study, the 2021 emission measurement results of a cement production facility located in Düzce Province were tested against emission concentrations calculated using the facility's production capacity and process-based emission factor values for the same year, as defined by the Ministry of Environment, Urbanization and Climate Change. The analysis focused on Particulate Matter (PM) and Nitrogen Oxides (NO_x) parameters, evaluating both theoretical (calculation-based) and practical (measurement-based) concentration values. The primary objective was to compare the agreement between these two approaches. Based on the field measurements conducted under the guidance of the Ministry, the average annual concentrations of PM and NO_x (NO + NO₂) were 2.64 µg/m³ and 436.64 µg/m³, respectively. In contrast, the calculated values based on the emission factor (EF) with 2021 production data were 13.12 µg/m³ for PM and 157.61 µg/m³ for NO_x. These discrepancies are influenced by several factors, including operational duration, production capacity during measurement periods, and other activities occurring at the facility. The obtained concentrations of emissions from both methods were then simulated with the AERMOD Air Quality Dispersion Model and using 2021 meteorological data, and spatial map plots of the resulting concentrations were made to observe the impact in the surrounding region.

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INTRODUCTION

Environmental problems around the world are becoming a major threat due to increasing population, industrialization and ownership control. Especially since the 20th century, industrial developments in the world have increased gradually, and in parallel with this increase, environmental problems, especially air pollution, have become unavoidable. There are various institutions and organizations in the world to minimize and control air pollution and to ensure compliance with these values by determining limit values for pollutants and to monitor compliance. For example, these limit values are determined and monitored by many institutions and organizations such as the United States Environmental Protection Agency (USEPA), the World Health Organization (WHO) and the Ministry of Environment, Urbanization and Climate Change (MoEU) in Türkiye [1].

Different air pollution explanations are available. According to the definition made by the Ministry of Environment, Urbanization and Climate Change, air pollution is the situation where air pollutant parameters exceed the concentrations determined normally, adversely affecting the

health of living beings and/or causing material damage [2]. According to Müezzinoğlu [3], air pollution is the presence of one or more types of pollutants in the open air (outside buildings) at certain concentrations and over certain periods of time, in amounts that harm human, plant and animal life, commercial or personal belongings, ecological balance and environmental quality. Air pollution is divided into four groups according to its sources. These are air pollution caused by i) heating, ii) motor vehicles, iii) industry, and iv) natural sources. Some of the prominent air pollutants formed by these sources are; sulfur dioxide (SO₂), particulate matter (PM₁₀ and PM_{2.5}), nitrogen dioxide (NO₂), carbon monoxide (CO), volatile organic compounds (VOCs), and heavy metals [4]. Air quality management is assessed by the Ministry of Environment, Urbanization and Climate Change according to the limit values of various regulations such as the Industrial Air Pollution Control Regulation (SKHKKY) and the Air Quality Assessment and Management Regulation (HKDYY). The purpose of the Industrial Air Pollution Control Regulation (SKHKKY) is to "control the emissions in the form of soot, smoke, dust, gas, steam and aerosol released into the atmosphere as a result of the activities of industrial and energy production facilities; to protect people and their

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environment from the dangers that may arise from pollution in the air receiving environment; to eliminate the negative effects that cause significant damage to the public and neighborly relations arising from air pollution in the environment and to determine the procedures and principles for preventing the emergence of these effects" [5]. The purpose of the Air Quality Assessment and Management Regulation (AQAM) is to "define and establish and evaluate air quality targets to prevent or reduce the harmful effects of air pollution on the environment and human health, and to maintain the current situation where air quality is good and improve it in other cases" [6].

Cement, which has become indispensable with growing populations and advancing technology, plays a vital role in industrial development. Manufacturing processes of cement factories are also a fundamental source of air pollution. Therefore, as with most industrial sectors, AERMOD air quality dispersion modeling is applied to determine the area affected by emissions, simulate concentration levels, and assess potential environmental impacts. In Türkiye, use of the AERMOD dispersion model is regulated by Regulation on the Control of Industrial Air Pollution [5]. Article 6 (paragraphs e and g) provides that installations with mass flow rates exceeding the given limits in Annex-2 are required to evaluate their effect on air pollution via an internationally accepted dispersion model. In this context, AERMOD, being one of the globally acknowledged Gaussian dispersion models, is very commonly used and recognized by the Ministry of Environment, Urbanization and Climate Change.

There are many air quality modeling studies in the literature that have been carried out using the AERMOD air quality dispersion model as well as other air quality models. In a study conducted by Afzali et al. [7] in the Pasir Gudang Industrial Zone in Johor, Malaysia, the distributions of SO_2 , NO_2 , and PM_{10} concentrations were determined using the AERMOD dispersion model coupled with WRF (Weather Research and Forecasting). In another study conducted by Kawashima et al. [8], the emission inventory of a total of 16 refining units, 1,730 thermal power plants, 96 cement industries, and 64 paper and pulp industries—comprising stationary pollutant sources in Brazil—was established. The results showed annual emission values of 857,000 tons for NO_x , 123,000 tons for SO_x , 212,000 tons for CO, 104,000 tons for PM, 114,000 tons for TOC (total organic carbon), and 476,000 tons for CO_2 [8]. In the study conducted by Taygun et al. [9], the distributions of PM_{10} , SO_2 , NO_x , and CO concentrations were simulated using the AERMOD air quality dispersion model over a 140 km × 110 km area including industrial zones, residential areas, and traffic sources in Kütahya [9]. In the Samsun Organized Industrial Zone, Akdemir [4] conducted a modeling study using the AERMOD air quality model to determine SO_x and NO_x concentrations. In another study conducted in Denizli Province, the distributions of NO_2 , SO_2 , and PM_{10} emissions originating from the use of coal and natural gas in industrial facilities located in the Denizli Organized Industrial Zone (DOSB) were determined under three different scenarios using the AERMOD dispersion model [10]. Scenario 1 represented conditions without air pollution control systems, Scenario 2 represented conditions where all industrial facilities used natural gas, and Scenario 3 represented conditions where air pollution control systems were applied. According to Scenario 1, the highest annual average concentrations of NO_2 , SO_2 , and PM_{10} were determined as 39 $\mu\text{g}/\text{m}^3$, 176 $\mu\text{g}/\text{m}^3$, and 291 $\mu\text{g}/\text{m}^3$, respectively. According to Scenario 2, the highest annual average concentrations were 32 $\mu\text{g}/\text{m}^3$, 0.07 $\mu\text{g}/\text{m}^3$, and 0.9 $\mu\text{g}/\text{m}^3$, respectively. According to Scenario 3, the

highest annual average concentrations were observed as 24 $\mu\text{g}/\text{m}^3$, 88 $\mu\text{g}/\text{m}^3$, and 3.8 $\mu\text{g}/\text{m}^3$, respectively [10].

This study aims to evaluate and compare pollutant concentration values measured at a cement production facility in Düzce Province with those calculated using emission factors and simulated through the AERMOD air quality dispersion model. The novelty of this study lies in the dual approach applied: both direct stack measurement and emission factor-based calculation were used together for a cement plant in Düzce, providing a unique comparative framework. This work therefore contributes to the literature by demonstrating result of the methodological differences and discussing possible reasons, and proposing Tier-3 based emission factor development.

MATERIALS AND METHODS

Study Area

Düzce is a province located within the Western Black Sea borders of Turkey. Düzce Province is located between 40° - 42° north latitude and 30° - 33° east longitude. Düzce's population and population growth rate between 2019-2023 are given in Table 1 according to the information obtained from TURKSTAT [11]. Air pollution in Düzce changes and progresses depending on the heating, transportation and industrial sectors and meteorological factors. The number of houses increases due to the population growth and the types and amounts of fuel used in heating in houses differ due to the differences in fuel types and systems.

Table 1. Population and population growth rate of Düzce Province between 2019-2023 [11]

Year	Total	Population Growth Rate
2023	409.865	11,6
2022	405.131	10,3
2021	400.976	13,3
2020	395.679	8,9
2019	392.166	11,1

According to the data obtained from the company data of 2021, the facility is located on a 720,000 m² surface area and a closed area of 34,893 m². The satellite image of the facility is given in Figure 1.



Figure 1. Satellite image of the location of the facility under study

In line with the information and data obtained from the facility's 2022 final EIA report, the facility's production capacity for 2021 was determined as 1,998,720,000 kilograms, and the working time was determined as 12 months, 330 days, 24 hours a day over 3 shifts. The facility has a closed clinker stock hall with a capacity of 120,000 tons, according to the facility's capacity report, the clinker production capacity is 1,500,000 tons/year and the 15-day clinker production capacity is calculated as 75,000 tons. Cement production in the facility is carried out on an area of 7,232 m².

Table 2 shows the coordinates, heights, diameters, stack gas velocities, and flow rate of the chimneys where the measurements were carried out in the facility.

According to the Industrial Air Pollution Control Regulation [5], the minimum chimney height for large-scale industrial facilities is 30 m. In Table 2, Chimney 1 has a height of 100 m and Chimney 2 has a height of 57 m, both of which are above the minimum legal requirement. Flue gas exit velocities also

comply with the regulation (minimum 4 m/s). Therefore, the chimney parameters of the studied facility are in compliance with the current Turkish legislation.

Emission Calculations

Since the study was conducted on emissions originating from a cement production facility in Düzce province, the process related to cement production stages is shown in Figure 2 below.

In this study, the mass flow rates were first determined using 2021 measurement data at the factory's two source chimneys, where PM, NO, and NO₂ parameters were measured. These values of PM, NO, and NO₂ parameters were used to calculate and estimate the emission concentrations. The results were then simulated by using the AERMOD Air Quality Dispersion Model. Table 3 presents the mass flow rates obtained from the measurements of the analyzed chimneys.

Table 2. Source chimney information

Source Chimney Name	Measured Parameters	Source Chimney Coordinates	Chimney Height (meters)	Chimney Diameter (meters)	Flue Gas Velocity (m/s)	Chimney Temperature (°C)	Flue Gas Flow Rate (m ³ /hour)
Chimney 1	Particulate Matter	40.9444	100	4.25	8.40	100.87	428992.33
	NO	31.3948					
	NO ₂						
Chimney 2	Particulate Matter	40.9443	57	1.80	6.60	66.33	60461.84
	NO	31.3948					
	NO ₂						

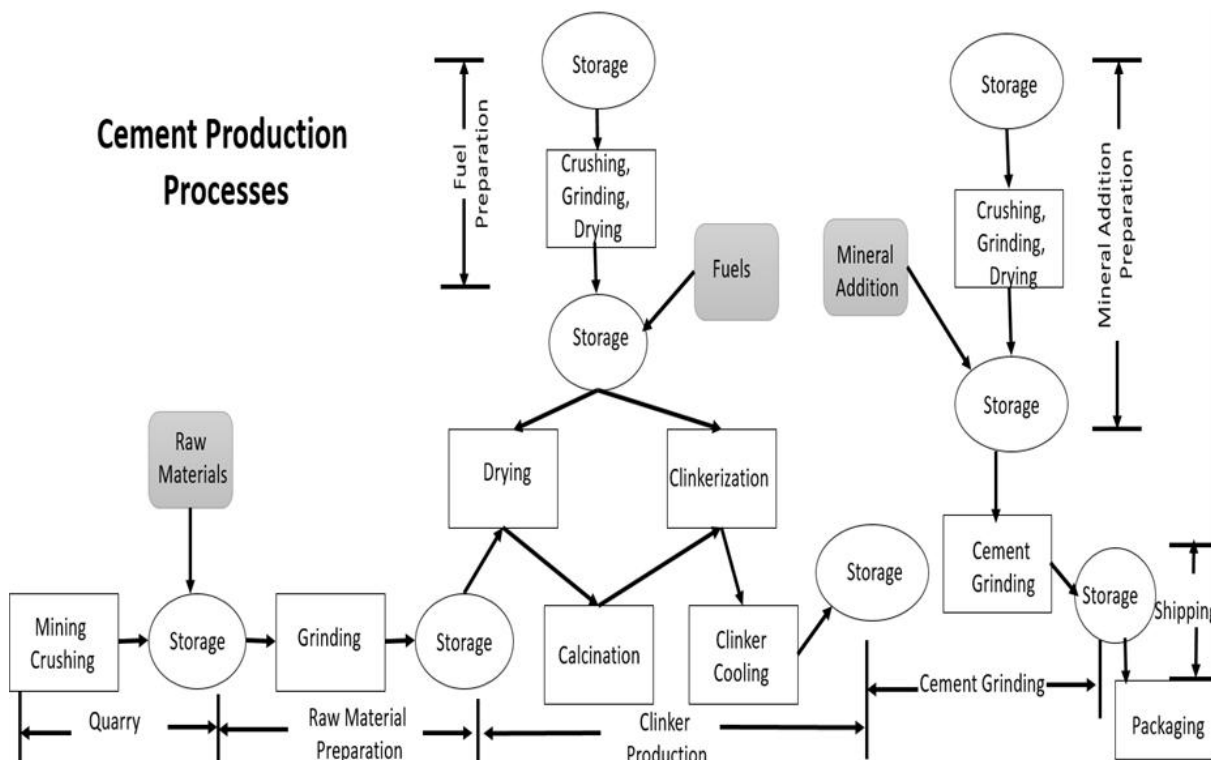


Figure 2. Cement production processes [12]

Table 3. Mass flow rate calculation results of source chimneys

Chimney 1		
Parameter	Mass Flow (kg/hour)	Mass Flow (g/s)
Particulate Matter	0.3678	0.1021
NO	38.861	10.7947
NO ₂	60.0439	16.6788
Chimney 2		
Parameter	Mass Flow (kg/hour)	Mass Flow (g/s)
Particulate Matter	3.0605	0.8501
NO	196.55	54.5972
NO ₂	305.4436	84.8454

Table 4. Emission factor values for Particulate Matter and NO_x in cement industry [13]

Parameter	Emission Factor (EF) (mg/Nm ³)
Particulate Matter	25
NO _x	300

The emission factor (EF) is an indicative value that quantifies the quantity (mass) of air pollutants emitted to the atmosphere in terms of a defined emission source. The emission factors are used to estimate emissions from a defined process, facility, or area and to forecast their potential

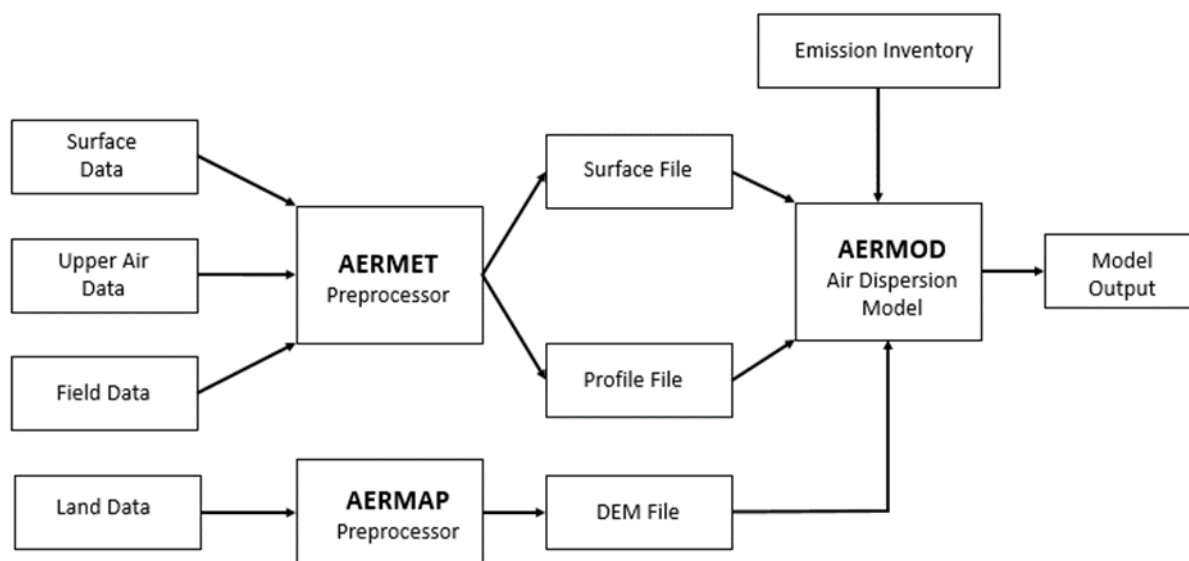
impact on air quality. They are particularly valuable for developing emission estimates in cases where direct measurements are not feasible. Processes within the plant responsible for Particulate Matter (PM) and Nitrogen Oxides (NO_x) emissions were identified first. The corresponding emission factor values for these processes were obtained from the EMEP/Corinair Emission Inventory Guidebook [13]. The emission factor values applied in the study are presented in Table 4. Using the facility's daily production capacity and the EFs, the emission values for PM and NO_x parameters were calculated, and the findings presented in Table 5.

Dispersion Modeling

The AERMOD (American Meteorological Society/Environmental Protection Agency Regulatory Model) air quality model was used to determine the dispersion of the concentration results obtained in the study. The AERMOD air quality model is among the most preferred dispersion models that is used in air quality assessments. In addition to identifying and evaluating the dispersion of pollutant emissions (e.g., CO, NO_x, SO₂, VOCs) from industrial sources under existing meteorological conditions, the model also estimates ground-level pollutant concentrations by integrating source emission data with meteorological parameters and surface characteristics [14]. The model can predict both short-term (instantaneous) and long-term (average) concentrations. The AERMOD model requires emissions and meteorological input data to generate pollutant concentrations in the emission inventory. The AERMOD model consists of three modules (Figure 3). These modules are AERMET, AERMAP and AERMOD.

Table 5. Mass flow rates calculated using daily production capacity and EF for Particulate Matter and NO_x parameters in the facility under study

Parameter	Capacity (ton/day) x EF (kg/ton)	Mass Flow (g/s)
Particulate Matter	$((6000 \text{ tons clinker}) \times (25 \text{ mg/Nm}^3) \times (2300 \text{ m}^3/\text{ton clinker})) / (1000 \times 86400)$	3.99
NO _x	$((6000 \text{ tons clinker}) \times (300 \text{ mg/Nm}^3) \times (2300 \text{ m}^3/\text{ton clinker})) / (1000 \times 86400)$	47.91

**Figure 3.** Data flow of AERMOD [15]

The 2021 meteorological station data of Düzce Province were obtained from the General Directorate of Meteorology. Hourly ambient temperature, cloud cover, wind speed and wind direction were obtained from the surface weather station, while soundings were collected from the rawinsonde station twice a day. As a result of the measurements taken at the facility, emission concentrations were obtained using the mass flow rates and emission factors calculated with the mass flow values calculated using AERMOD air quality modeling, and the obtained emission concentrations were visualized with maps.

Topographic data related to the study area terrain elevations were extracted from the United States Geological Survey Digital Elevation Map. In the model, a 10 km × 10 km Cartesian grid system was utilized. The determined grid system area was generated with 250 m intervals. As a result of AERMOD Air Quality modeling, concentrations at receptor points were used to obtain daily, monthly, and annual concentration distribution maps.

The wind rose was formed using the WRPLOT View program provided by Lakes Environmental using the wind data for Düzce province for 2021 (Figure 4). The wind rose includes both wind speed and wind direction distribution. The predominant wind direction is the southwest.

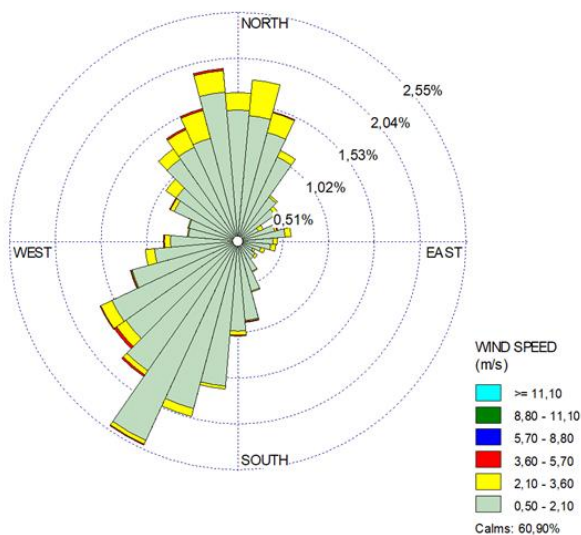


Figure 4. 2021 Düzce province wind rose

RESULTS AND DISCUSSION

In the study, two different emission calculations were carried out for a cement plant operating in Düzce Province. In the first methodology, the stack gas measurements were used for calculating the mass emission rates of PM, NO, and NO₂. On the second methodology, available emission factors were used to estimate the mass emissions [13]. The concentrations were gathered from the stacks in 2021. Therefore, modelling study was performed for the year of 2021. The ground-level concentrations were estimated in $\mu\text{g}/\text{m}^3$.

The annual average PM concentration for Chimney 1 and Chimney 2 emissions in 2021 was calculated as $2.64 \mu\text{g}/\text{m}^3$ at coordinates X: 365000.00, Y: 4533329.00. The concentration distribution is shown in Figure 5. This value shows the possible highest average PM concentration around the cement factory.

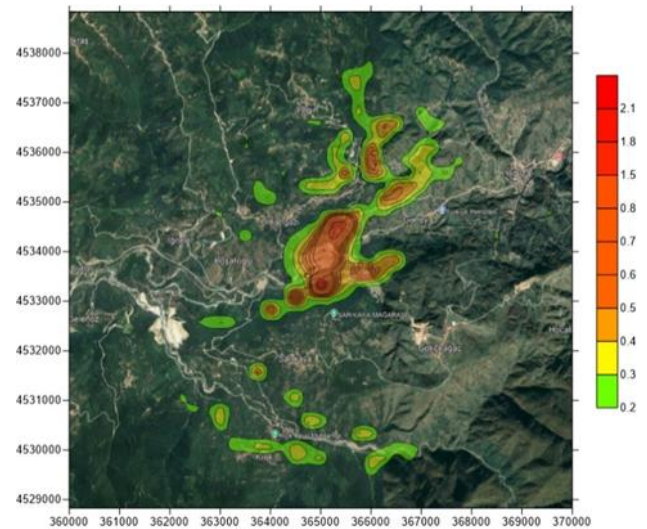


Figure 5. Annual average PM concentration (in $\mu\text{g}/\text{m}^3$) distribution map for Chimney 1 and Chimney 2

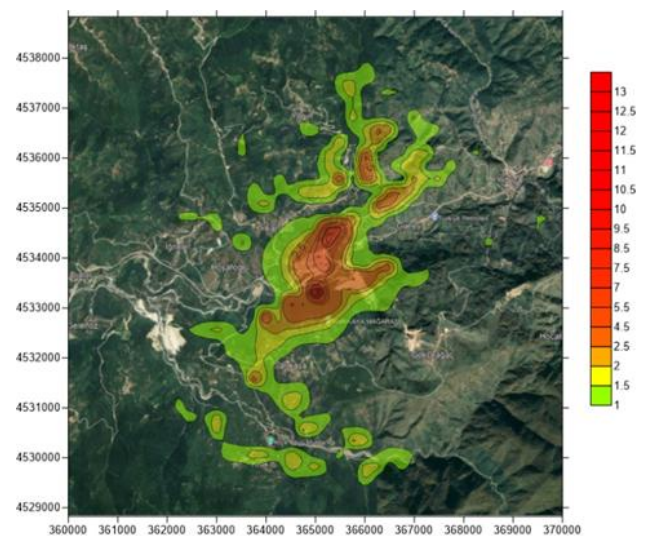


Figure 6. Annual average PM (in $\mu\text{g}/\text{m}^3$) concentration distribution map according to EF calculation result

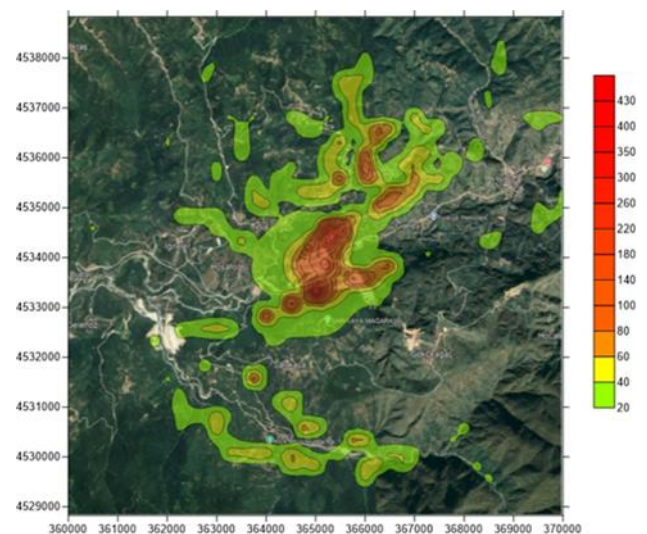


Figure 7. Annual average NO_x concentration (in $\mu\text{g}/\text{m}^3$) distribution map for Chimney 1 and Chimney 2

As a result of the calculations made with the EF in line with the data in 2021, the annual average Particulate Matter concentration was calculated as $13.12 \mu\text{g}/\text{m}^3$ at coordinates X: 365000.00, Y: 4533329.00. The measured concentration distribution is shown in Figure 6.

The estimated ground-level values were lower for the simulation which was prepared using the stack gas measurements.

The annual average NO_x ($\text{NO}+\text{NO}_2$) concentration for Chimney 1 and Chimney 2 in 2021 was measured as $436.64 \mu\text{g}/\text{m}^3$ at coordinates X: 365000.00, Y: 4533329.00. The measured concentration distribution is shown in Figure 7.

As a result of the calculations made with the EF calculations, the annual average NO_x concentration was calculated as $157.61 \mu\text{g}/\text{m}^3$ at coordinates X: 365000.00, Y: 4533329.00, where it had the peak value. The measured concentration distribution is shown in Figure 8.

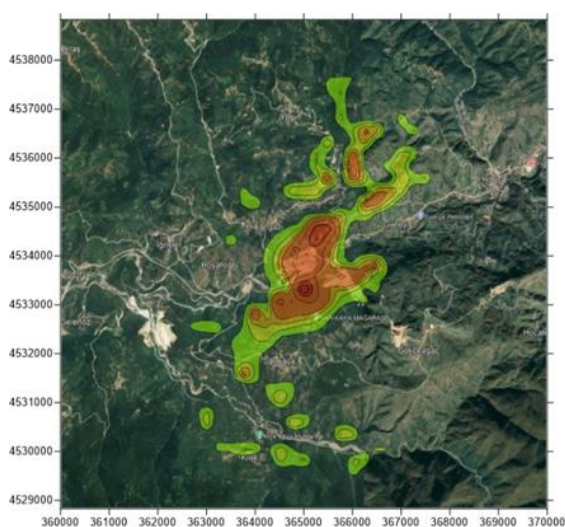


Figure 8. Annual average NO_x concentration (in $\mu\text{g}/\text{m}^3$) distribution map according to EF calculation result

The stack emissions were higher for PM when EMEP/Corinair emission factors were used. However, higher NO_x emissions were calculated by using the stack measurements. These results indicated that alone emission factor or measurement is not reliable for each factory. Since different parameters were higher for emission factor-based calculation and measurement-based emission estimation. In this case, ambient air measurement seems mandatory for selecting the closest value.

The concentration difference at coordinates X: 365000.00, Y: 4533329.00, which is the difference between the annual average NO_x emissions was found to be $391.48 \mu\text{g}/\text{m}^3$. Figure 9 shows the concentration value distribution resulting from the differences obtained.

The concentration difference at coordinates X: 365000.00, Y: 4533329.00, which is the difference between the annual average PM concentration value measured in the chimney in 2021 and the annual average PM concentration value calculated as a result of the EF, was found to be $12.61 \mu\text{g}/\text{m}^3$. Figure 10 shows the concentration value distribution resulting from the differences.

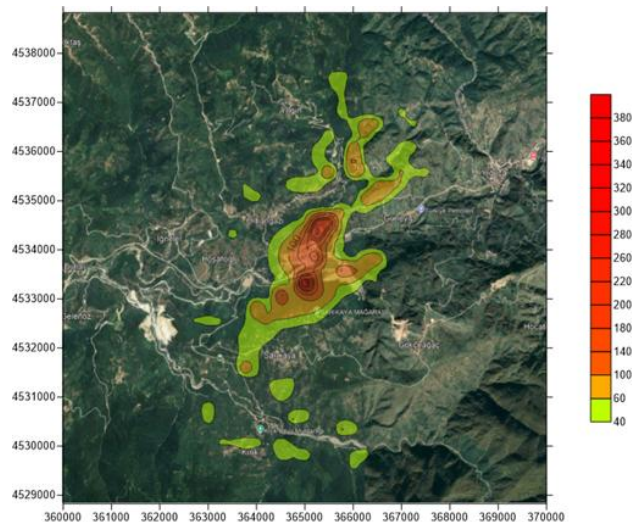


Figure 9. Difference map between the annual average NO_x concentration value measured in the chimney and the annual average NO_x concentration value according to the EF calculation result (in $\mu\text{g}/\text{m}^3$)

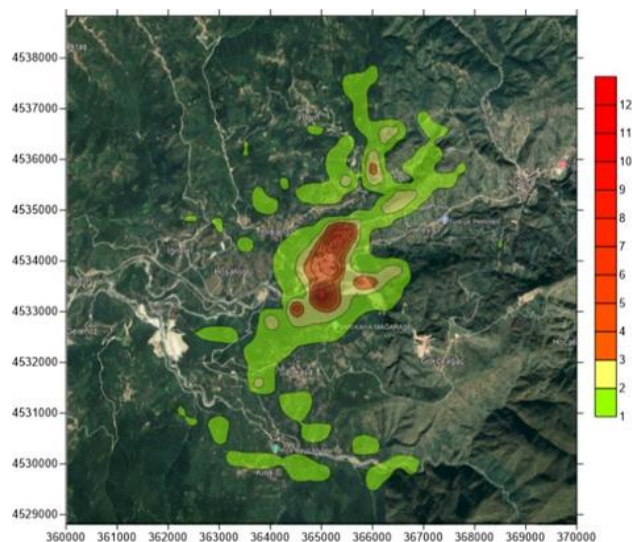


Figure 10. Difference map between the annual average PM concentration value measured in the chimney and the annual average PM concentration value according to the EF calculation result (in $\mu\text{g}/\text{m}^3$)

Comparing the measurement-based to emission factor (EF)-based model values showed considerable discrepancies. For particulate matter (PM), EF-based gave a yearly average concentration of $13.12 \mu\text{g}/\text{m}^3$ and measurement-based estimation was $2.64 \mu\text{g}/\text{m}^3$ with a positive bias of $10.48 \mu\text{g}/\text{m}^3$. This corresponds to an NMB of +397%, i.e., the EF-based method strongly overestimated PM concentrations when compared to direct stack measurements. On the contrary, in the case of nitrogen oxides (NO_x), the EF-based result ($157.61 \mu\text{g}/\text{m}^3$) was significantly smaller compared to the measurement-based one ($436.64 \mu\text{g}/\text{m}^3$), and the bias gained was $-279.03 \mu\text{g}/\text{m}^3$ with an NMB of -63.9%. This reveals a severe underestimation of NO_x emissions by the EF methodology.

The ambient PM concentrations were comparable to those reported from cement manufacturing plants in Iran [16] and Egypt [17]. The reported highest total suspended particulate matter concentration from cement factory emission was 2.31

$\mu\text{g}/\text{m}^3$ in Iran [16]. The highest annual average PM_{10} concentration around a cement manufacturing plant in Egypt was $9 \mu\text{g}/\text{m}^3$ [17].

Less annual average NO_x concentrations from cement factory were reported in Iran [16], Egypt [17] and India [18]. However, a study calculated NO_x emissions from cement factories in Thailand [19] was comparable with the results of this study. In Egypt, the highest annual NO_x concentration was $1.87 \mu\text{g}/\text{m}^3$ [17]. $14.92 \mu\text{g}/\text{m}^3$ annual NO_x concentrations were estimated in India [18]. Cement factory borne NO_x emissions were modelled to estimate ground-level concentrations in Thailand, where four different factories are present [19].

CONCLUSIONS

This study evaluated the annual average concentrations of PM and NO_x from two stacks of a cement factory. Two different emission calculations were conducted in the study. One exploited the measurement results, the other used the emission factors. The highest annual average PM concentration was found to be $2.64 \mu\text{g}/\text{m}^3$ according to stack measurement, whereas the corresponding value calculated using EF was $13.12 \mu\text{g}/\text{m}^3$. For NO_x , the measured annual average concentration was $436.64 \mu\text{g}/\text{m}^3$, while the EF-based calculation yielded a value of $157.61 \mu\text{g}/\text{m}^3$. The difference between the measured and calculated NO_x concentrations was $391.48 \mu\text{g}/\text{m}^3$, and the difference in PM concentrations was $12.61 \mu\text{g}/\text{m}^3$. These variations can be attributed to several influencing factors, including the daily, monthly, and annual operational capacities of the facility, as well as working conditions, hours of operation, and the types of raw materials used. It is also important to note that the facility was operating at full capacity during the measurement period, which may have influenced the concentration values obtained. In order to ensure that these concentrations comply with the limit values set out in relevant legislation, appropriate control measures must be implemented by both facility operators and authorized regulatory bodies. Ground-level concentration measurement can be used to validate the model results. On the other hand, further studies can be implemented to produce emission factors in Tier-3 level, which can help produce closer results to measured values.

DATA AVAILABILITY STATEMENT

The authors confirm that the data that supports the findings of this study are available within the article. Raw data that support the finding of this study are available from the corresponding author, upon reasonable request.

CONFLICT OF INTEREST

The authors declare that there is no conflict of interest with any individual, institution, or organization in the preparation, evaluation, or publication of this study.

USE OF AI FOR WRITING ASSISTANCE

Not declared.

ETHICS

There are no ethical issues with the publication of this manuscript.

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